

Pie Charts

The Journey from Data to Visual Intelligence

Pie Charts Masterclass

1. What is a Pie Chart?

A pie chart shows how different categories share a whole.

Key Idea

Pie Chart = Parts of a Whole

Example

If a pizza has 8 slices and you eat 4 slices:

$$4 / 8 = 50\%$$

You ate half of the pizza.

2. Why Do We Use Pie Charts?

Pie charts help us:

- Show percentages
- Show proportions
- Understand how a whole is divided

Diagram

Whole
↓
Parts
↓
Pie Chart

3. Example: Favourite Subjects

Subject	Students
Science	15
Maths	10
English	5

Total Students

$$15 + 10 + 5 = 30$$

Percentages

Science = 50%
Maths = 33%
English = 17%

4. Bar Chart vs Pie Chart

Bar Chart

- Compares values
- Shows which category is bigger

Pie Chart

- Shows percentages

- Shows how categories share a whole

Quick Rule

Bar Chart → Comparison

Pie Chart → Composition

5. When to Use a Pie Chart?

Use a pie chart when:

- Total matters
- Categories form one whole
- Percentages are important

Examples

- Budget spending
- Favourite subjects
- Screen time usage
- Mobile usage

6. When NOT to Use a Pie Chart?

Avoid pie charts when:

- There are too many categories
- Small differences must be compared

Example

Apple	50
Mango	49
Banana	48
Orange	47

Grapes	46
Guava	45

Too many slices make the chart difficult to read.

Better Choice

Bar Chart

7. Rule of Thumb

Use Pie Charts

3–5 Categories

Use Bar Charts

More than 6 Categories

8. Understanding Percentages

Example

10 chocolates

Eat 5 chocolates

$$5 / 10 = 50\%$$

Eat 2 chocolates

$$2 / 10 = 20\%$$

Pie charts are based on percentages.

9. Pie Chart Detective Rule

All slices together must equal:

100%

Diagram

20%
30%
50%

100%

If the total is not 100%, something is wrong.

10. The 3D Pie Chart Problem

3D pie charts can make slices look bigger or smaller than they really are.

Remember

3D Pie Charts
Look Good
But Can Be Misleading

11. Pie Chart vs Bar Chart

Bar Chart	Pie Chart
Compare values	Show percentages
Best for ranking	Best for proportions
Works with many categories	Best with few categories

Bar Chart	Pie Chart
More accurate comparison	Easier to see shares

12. Real-World Uses

Pie charts are used to show:

- Budget distribution
- Market share
- Device usage
- Customer age groups
- Screen time breakdown

13. Summary

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Data
↓
Percentages
↓
Parts of a Whole
↓
Pie Chart
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Key Points

- Pie charts show parts of a whole
- All slices must add up to 100%
- Best for percentages and proportions
- Use 3–5 categories for clarity
- Use bar charts when there are many categories

Remember

Bar Chart = Compare Categories

Pie Chart = Show Shares of a Whole